Karlsruhe Institute of Technology

IAM-AWP / Dep. of Metallic Materials/
Atomistic Modeling and Validation Group

From Atomic Jumps to Stress-Driven Hydrogen Transport

Subheadline: Arial, Regular, 50 pt, Black or Grey (70% Black)

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Missing layer in multiscale materials modelling workflow

The computational limit of numerical random-walks

The Status Quo:

Determining **diffusion coefficient** usually relies on **computationally heavy numerical simulations** (like KMC or MD) that average random-walk trajectories over massive time scales

Endless Computing Loop

The Inefficiency: Every change in temperature, strain, or migration barrier requires running the entire numerical simulation from scratch.

Bravais vs non-Bravais Lattices

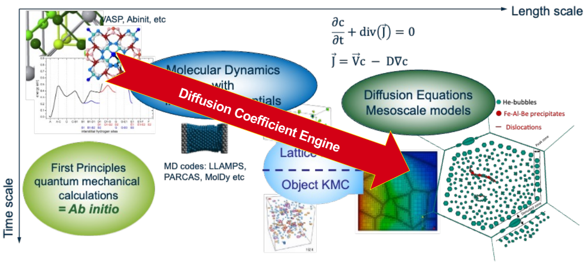
- Diffusion of vacancies or interstitials on Bravais lattice (with one site per unit cell) is **uncorrelated**. There is a simple analytical expression for calculating *DC*:

$$D_{ij}^{uncor} = \frac{1}{2}\sum_{\alpha=1}^{n}(h_i^\alpha \cdot h_j^\alpha)\, W(\vec{h}^\alpha)$$

- Diffusion on non-Bravais lattices (with multiple sites per unit cell) is **mutually correlated** and requires solving a master equation:

$$\frac{dP_\alpha(\boldsymbol{l},t)}{dt} = \sum_{\beta=1}^{N}\sum_{\boldsymbol{l}'}[W_{\beta\alpha}^{\boldsymbol{l}'\boldsymbol{l}} P_\beta(\boldsymbol{l}',t) - W_{\alpha\beta}^{\boldsymbol{l}\boldsymbol{l}'} P_\alpha(\boldsymbol{l},t)]$$

Bridging the gap in multiscale modelling workflow



Automated evaluation of analytical transport coefficients

Analytical Transport Parameters:

Diffusion tensors and drift velocities from atomistic data

- **Diffusion tensor** is completely defined via its principal components, i.e., diffusion along principal axis.
- Ishioka and Koiwa reduced the 3D-diffusion to a 1D problem by projecting to a crystal direction perpendicular to a family of crystal planes.
- They obtained very compact representation of $D_{\xi\xi}$:

$$D_{\xi\xi} = -\frac{1}{2}(\mathrm{d}_\xi)^2\left[\frac{\mathrm{d}^2}{d\theta^2}det\left(\hat{A}(\theta)\right)\right]_{\theta=0}/det(\hat{B})$$

where $\hat{A}(\theta)$ is the Fourier transform of the transition matrix and $\theta = (\vec{a}_1\vec{\xi})t = d_\xi \cdot t$, where d_ξ is the interplanar distance.

$\hat{B}$ is $\hat{A}(\theta = 0)$ in which one row is replaced with ones.

- **Drift velocity** appears in transport equation if **external field** is applied.
- In this case, jumps **along** and **against** the external force are **not equivalent**
- **Drift velocity** can be calculated similarly to the diffusion coefficient

$$\mathrm{v_{drift}} = -i\frac{\partial}{\partial\theta}\det(\boldsymbol{A}(\theta))\Big|_{\theta=-i\beta\mathrm{F}}/\det(\boldsymbol{B})$$

- If external field can be treated as perturbation $\frac{\mathrm{F}}{\mathrm{k_B T}} \ll 1$,

Einstein-Smoluchowski relation for mobility μ is restored

$$v_{drift} = \mu F \quad \mu = \frac{D}{\mathrm{k_B T}}$$

Resurrecting and automation of the Ishioka-Koiwa method

Energetics:

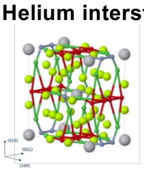
- Input crystal structure → Find candidate interstitial sites
- Find candidate interstitial sites → Perform static *ab initio* calculations ⇒ site solution energies
- Perform static *ab initio* calculations ⇒ site solution energies → Perform *ab initio* dimer calculations => intra-sites migration energies
- Perform *ab initio* dimer calculations => intra-sites migration energies → Set of solution and saddle point energies
- Set of solution and saddle point energies → **transport_coefficient_evaluator**:
  - performs numerical evaluation of DC using *ab initio* data

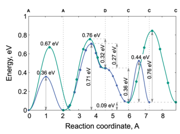
Geometry & Topology

Python package **SymDCT**

- **jump_network_builder**:
  - constructs jumpnetwork
- **matrix_builder**:
  - constructs matrix $\hat{A}(\theta)$
  - reduce its size using crystal symmetries
- **transport_coefficient_solver**:
  - calculates analytical expression for the DC along given direction

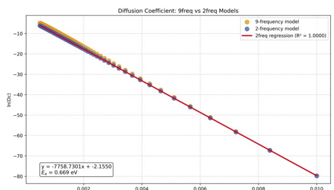
Helium interstitial diffusion in $Be_{12}Ti$





Exact solution vs 2-freq model

$$D_{<001>} = \left(\frac{c}{2}\right)^2\frac{num}{denom} \qquad D_{<001>} = \frac{1}{2}\left(\frac{c}{2}\right)^2\frac{\mathrm{W_1 W_2}}{\mathrm{W_1 + W_2}}$$



Shifting from numerical averaging to analytical precision

	Numerical Simulations	SymDCT Analytical Approach
Methodology	Stochastic averaging over time and space	Exact analytical expressions
Computational Expense	Hight: Requires rerun for every parameter change	Near-zero: Evaluates the generated function instantly
Reusability	Specific to one rigid condition	Universal for any material with identical jump-network topology
Complex Conditions	Struggles with inhomogeneous fields	Natively supports liner-response analysis for external fields

SymEngine

Link to GitHub repository

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